

A proof of the conjectured linear recurrence and generating function for OEIS A256814

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1 The conjecture

The Online Encyclopedia of Integer Sequences records [A256814](#) as

Number of length $n+6$ 0..1 arrays with at most two downsteps in every 6 consecutive neighbor pairs

and carries in its Formula section the lines:

Empirical: $a(n) = 2a(n-1) - a(n-2) + 2a(n-3) - a(n-4) + 3a(n-6) - 2a(n-7) - 6a(n-9) + 4a(n-10).$

Empirical g.f.: $x(120 - 11x + 104x^2 - 39x^3 + 48x^4 + 93x^5 - 190x^6 - 128x^7 - 250x^8 + 252x^9) / ((1 - x)(1 - x - 2x^3 - x^4 - x^5 - 4x^6 - 2x^7 - 2x^8 + 4x^9)).$ - Colin Barker, Jan 24 2018

That line carries no separate signature; the entry itself is by R. H. Hardin (Apr 10 2015).

The line quoted ending in that signature is contributed by Colin Barker on Jan 24 2018.

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 7, last modified Jan 24 2018, the entry records no proof and links to none, so the statement is open. This note settles it.

Written out, the claim is

$$a(n) = 2a(n-1) - a(n-2) + 2a(n-3) - a(n-4) + 3a(n-6) - 2a(n-7) - 6a(n-9) + 4a(n-10).$$

stated without a range; it first has meaning at $n = 11$, the least index at which all of $a(n-1), \dots, a(n-10)$ are terms of the sequence.

Written out, the claim is

$$\sum_{n \geq 1} a(n) x^n = \frac{120x - 11x^2 + 104x^3 - 39x^4 + 48x^5 + 93x^6 - 190x^7 - 128x^8 - 250x^9 + 252x^{10}}{1 - 2x + x^2 - 2x^3 + x^4 - 3x^6 + 2x^7 + 6x^9 - 4x^{10}}.$$

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

The objects are the words $w_1 w_2 \cdots w_L$ over $\{0, 1, \dots, 1\}$ of length $L = n + 6$ satisfying the entry's conditions, which are, clause by clause:

- *at most two downsteps in every 6 consecutive neighbor pairs*

Each of these is a condition on a window of consecutive terms of bounded width, except the labelling clause where present, which says that the first occurrences of the values happen in increasing order.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

Read the word one letter at a time. Every condition is decided by a window of at most a fixed number of consecutive letters, so it becomes checkable as soon as that many letters have been read; and the labelling clause, where the entry has one, is decided by how many distinct values have been introduced so far, a number between 0 and 2.

Take as state the last few letters together with that counter. Appending a letter is a transition; a word of length L is a walk of length L from the empty state, and every word arises from exactly one walk. With M the adjacency matrix of the state digraph, w the indicator of the start and f of the accepting states, $a(n) = w^\top M^L f$, so the count is C-finite.

4 The degree bound, derived

The reachable part of that digraph has 127 states, 127 after trimming. Identifying states with the same number of completions of every length, and then the mirror identification on the edges coming in, leaves $S = 19$ blocks. The count satisfies the linear recurrence given by the characteristic polynomial of the 19×19 quotient matrix; that degree bound is computed here, not assumed.

5 The decision procedure

Let $c_1, \dots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^D c_i a(n-i)$, let $q(t) = t^D - \sum_{i=1}^D c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m = \tilde{w}^\top L^{m-1} q(L) \tilde{f} \quad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a , one gets $u_m = a(n) - \sum_{i=1}^D c_i a(n-i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^\top L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L , of degree 19: writing that polynomial as $t^{19} - \sum_{i=1}^{19} \lambda_i t^{19-i}$ gives $u_{m+19} = \sum_{i=1}^{19} \lambda_i u_{m+19-i}$. Hence 19 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1}y$. The matrix M is never formed.

6 Results

Theorem 1. *For A256814, the recurrence $a(n) = 2a(n-1) - a(n-2) + 2a(n-3) - a(n-4) + 3a(n-6) - 2a(n-7) - 6a(n-9) + 4a(n-10)$ holds for every $n \geq 11$ — at every index at which it can be stated.*

Proof. The residuals u_m were computed exactly for $m = 1, \dots, 62$. Every one of them vanished. After it, more than $S = 19$ consecutive residuals vanish, so by Section 5 every later residual vanishes as well. $\square$

Theorem 2. *For A256814, the generating function is exactly $\sum_{n \geq 1} a(n)x^n = \frac{120x - 11x^2 + 104x^3 - 39x^4 + 48x^5 + 93x^6 - 190x^7 + 120x^8 - 11x^9 + 104x^{10} - 39x^{11} + 48x^{12} + 93x^{13} - 190x^{14} + 120x^{15} - 11x^{16} + 104x^{17} - 39x^{18} + 48x^{19} + 93x^{20} - 190x^{21} + 120x^{22} - 11x^{23} + 104x^{24} - 39x^{25} + 48x^{26} + 93x^{27} - 190x^{28} + 120x^{29} - 11x^{30} + 104x^{31} - 39x^{32} + 48x^{33} + 93x^{34} - 190x^{35} + 120x^{36} - 11x^{37} + 104x^{38} - 39x^{39} + 48x^{40} + 93x^{41} - 190x^{42} + 120x^{43} - 11x^{44} + 104x^{45} - 39x^{46} + 48x^{47} + 93x^{48} - 190x^{49} + 120x^{50} - 11x^{51} + 104x^{52} - 39x^{53} + 48x^{54} + 93x^{55} - 190x^{56} + 120x^{57} - 11x^{58} + 104x^{59} - 39x^{60} + 48x^{61} + 93x^{62} - 190x^{63} + 120x^{64} - 11x^{65} + 104x^{66} - 39x^{67} + 48x^{68} + 93x^{69} - 190x^{70} + 120x^{71} - 11x^{72} + 104x^{73} - 39x^{74} + 48x^{75} + 93x^{76} - 190x^{77} + 120x^{78} - 11x^{79} + 104x^{80} - 39x^{81} + 48x^{82} + 93x^{83} - 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11x^{821} + 104x^{822} - 39x^{823} + 48x^{824} + 93x^{825} - 190x^{826} + 120x^{827} - 11x^{828} + 104x^{829} - 39x^{830} + 48x^{831} + 93x^{832} - 190x^{833} + 120x^{834} - 11x^{835} + 104x^{836} - 39x^{837} + 48x^{838} + 93x^{839} - 190x^{840} + 120x^{841} - 11x^{842} + 104x^{843} - 39x^{844} + 48x^{845} + 93x^{846} - 190x^{847} + 120x^{848} - 11x^{849} + 104x^{850} - 39x^{851} + 48x^{852} + 93x^{853} - 190x^{854} + 120x^{855} - 11x^{856} + 104x^{857} - 39x^{858} + 48x^{859} + 93x^{860} - 190x^{861} + 120x^{862} - 11x^{863} + 104x^{864} - 39x^{865} + 48x^{866} + 93x^{867} - 190x^{868} + 120x^{869} - 11x^{870} + 104x^{871} - 39x^{872} + 48x^{873} + 93x^{874} - 190x^{875} + 120x^{876} - 11x^{877} + 104x^{878} - 39x^{879} + 48x^{880} + 93x^{881} - 190x^{882} + 120x^{883} - 11x^{884} + 104x^{885} - 39x^{886} + 48x^{887} + 93x^{888} - 190x^{889} + 120x^{890} - 11x^{891} + 104x^{892} - 39x^{893} + 48x^{894} + 93x^{895} - 190x^{896} + 120x^{897} - 11x^{898} + 104x^{899} - 39x^{900} + 48x^{901} + 93x^{902} - 190x^{903} + 120x^{904} - 11x^{905} + 104x^{906} - 39x^{907} + 48x^{908} + 93x^{909} - 190x^{910} + 120x^{911} - 11x^{912} + 104x^{913} - 39x^{914} + 48x^{915} + 93x^{916} - 190x^{917} + 120x^{918} - 11x^{919} + 104x^{920} - 39x^{921} + 48x^{922} + 93x^{923} - 190x^{924} + 120x^{925} - 11x^{926} + 104x^{927} - 39x^{928} + 48x^{929} + 93x^{930} - 190x^{931} + 120x^{932} - 11x^{933} + 104x^{934} - 39x^{935} + 48x^{936} + 93x^{937} - 190x^{938} + 120x^{939} - 11x^{940} + 104x^{941} - 39x^{942} + 48x^{943} + 93x^{944} - 190x^{945} + 120x^{946} - 11x^{947} + 104x^{948} - 39x^{949} + 48x^{950} + 93x^{951} - 190x^{952} + 120x^{953} - 11x^{954} + 104x^{955} - 39x^{956} + 48x^{957} + 93x^{958} - 190x^{959} + 120x^{960} - 11x^{961} + 104x^{962} - 39x^{963} + 48x^{964} + 93x^{965} - 190x^{966} + 120x^{967} - 11x^{968} + 104x^{969} - 39x^{970} + 48x^{971} + 93x^{972} - 190x^{973} + 120x^{974} - 11x^{975} + 104x^{976} - 39x^{977} + 48x^{978} + 93x^{979} - 190x^{980} + 120x^{981} - 11x^{982} + 104x^{983} - 39x^{984} + 48x^{985} + 93x^{986} - 190x^{987} + 120x^{988} - 11x^{989} + 104x^{990} - 39x^{991} + 48x^{992} + 93x^{993} - 190x^{994} + 120x^{995} - 11x^{996} + 104x^{997} - 39x^{998} + 48x^{999} + 93x^{1000} - 190x^{1001} + 120x^{1002} - 11x^{1003} + 104x^{1004} - 39x^{1005} + 48x^{1006} + 93x^{1007} - 190x^{1008} + 120x^{1009} - 11x^{1010} + 104x^{1011} - 39x^{1012} + 48x^{1013} + 93x^{1014} - 190x^{1015} + 120x^{1016} - 11x^{1017} + 104x^{1018} - 39x^{1019} + 48x^{1020} + 93x^{1021} - 190x^{1022} + 120x^{1023} - 11x^{1024} + 104x^{1025} - 39x^{1026} + 48x^{1027} + 93x^{1028} - 190x^{1029} + 120x^{1030} - 11x^{1031} + 104x^{1032} - 39x^{1033} + 48x^{1034} + 93x^{1035} - 190x^{1036} + 120x^{1037} - 11x^{1038} + 104x^{1039} - 39x^{1040} + 48x^{1041} + 93x^{1042} - 190x^{1043} + 120x^{1044} - 11x^{1045} + 104x^{1046} - 39x^{1047} + 48x^{1048} + 93x^{1049} - 190x^{1050} + 120x^{1051} - 11x^{1052} + 104x^{1053} - 39x^{1054} + 48x^{1055} + 93x^{1056} - 190x^{1057} + 120x^{1058} - 11x^{1059} + 104x^{1060} - 39x^{1061} + 48x^{1062} + 93x^{1063} - 190x^{1064} + 120x^{1065} - 11x^{1066} + 104x^{1067} - 39x^{1068} + 48x^{1069} + 93x^{1070} - 190x^{1071} + 120x^{1072} - 11x^{1073} + 104x^{1074} - 39x^{1075} + 48x^{1076} + 93x^{1077} - 190x^{1078} + 120x^{1079} - 11x^{1080} + 104x^{1081} - 39x^{1082} + 48x^{1083} + 93x^{1084} - 190x^{1085} + 120x^{1086} - 11x^{1087} + 104x^{1088} - 39x^{1089} + 48x^{1090} + 93x^{1091} - 190x^{1092} + 120x^{1093} - 11x^{1094} + 104x^{1095} - 39x^{1096} + 48x^{1097} + 93x^{1098} - 190x^{1099} + 120x^{1100} - 11x^{1101} + 104x^{1102} - 39x^{1103} + 48x^{1104} + 93x^{1105} - 190x^{1106} + 120x^{1107} - 11x^{1108} + 104x^{1109} - 39x^{1110} + 48x^{1111} + 93x^{1112} - 190x^{1113} + 120x^{1114} - 11x^{1115} + 104x^{1116} - 39x^{1117} + 48x^{1118} + 93x^{1119} - 190x^{1120} + 120x^{1121} - 1$*

8 References

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